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▶ Profit & Loss

Discount = MP − SP
Profit = SP − CP
Loss = CP − SP

$$\% = \frac{\text{Obtain}}{\text{Total}} \times 100$$

$$\text{Profit}\% = \frac{\text{Profit} \times 100}{CP}$$

$$D\% = \frac{D}{MP} \times 100$$

$$\text{Loss}\% = \frac{\text{Loss} \times 100}{CP}$$

$$\boxed{SP = CP\left[\frac{100 + P\%}{100}\right] \text{ or } CP\left[\frac{100 - L\%}{100}\right]}$$

$$SP = MP\left[\frac{100 - D\%}{100}\right]$$

Q: find cost of A which is sold at Rs 300 at loss of 25%

A:

$$\cancel{25 = \frac{75 \times 100}{CP}}$$

$$\cancel{CP = 300}$$

$$300 \times \frac{25}{100}$$

$$SP = CP\left[\frac{100 - L\%}{100}\right]$$

$$300 = CP\left[\frac{75}{100}\right]$$

$$400 = CP$$

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Q: CP of an artile = 560Rs. & A sells it at profit of 12.5%. find SP?

Ans. $SP = CP\left[\frac{100+12.5\%}{100}\right]$

$SP = \frac{560}{100} \times \frac{112.5}{100}$

$SP = \frac{112.5 \times 56}{100} = 630$

Q: A mobile costing Rs. 24000 is sold & a profit of 10% is recieved. find SP of mobile?

A. $SP = 24000 \times \frac{110}{100}$

$\Rightarrow 2400 \times 11$

Q: An iron is sold for Rs. 315 & incurred a loss of 10%. find cost price of iron?

A. $SP = CP\left[\frac{100-10\%}{100}\right]$

$315 = CP \times \frac{90}{100}$

$\frac{3150}{9} = CP$

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False Weight Problems

1. A sells item at CP but he uses a weight of 920 g for 1kg. find gain %?

Ans $\% = \frac{\text{obtain}}{\text{Total}} \times 100$ $1000 - 920 = 80\,g$

$P\% = \frac{P}{CP} \times 100 = \frac{80}{920} \times 100$

Q: In place of 1kg wt, a wt of 800g is used and goods are sold at CP, find profit %?

Ans $\frac{200}{800} \times 100 = 25\%$

Q: A dealer sell items at 20% Profit, & also uses a wt. which is 10% less than actual, find net profit %?

$\frac{30\%}{9\cancel{0}} \times \cancel{100} \Rightarrow \frac{300}{9} \Rightarrow 33.33\%$

Q: A shopkeeper is selling good at loss of 12.5% but he is using a wt. of 25g again actual wt. of 30g. find net profit or loss %?

Ans $[100 \pm P\%] \times \frac{\text{Act. wt}}{\text{Fals wt}} - 100$

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$$\left[\left[100-12.5\%\right] \times \frac{30}{25} - 100\right]$$

$$\left[\frac{87.5}{10} \times \frac{30}{25} - 100\right]$$

Q: An item is sold for Rs. 180 thereby suffering a loss of 10%. What should be SP if one wants 20% profit.

A

$200 \xrightarrow{-10\%} 180$

$\downarrow +20\%$

240

method

SP	P/L
180	16% L
S	20% P

$180 \times 26 = S \times 16$

$360 = S$

SP	P/L
180	90 → 10% L
S	120 → 20% P

$S \times \cancel{90} = \cancel{180} \times 120$

$S = 240$